Checkpoint 1 - Tumor Surface Target Prioritization

Cancer-driver-agnostic targeting of amplified surface proteins identifies MPZL1 for selective CAR-T cell therapy

For checkpoint 1, our group presents the "Cancer-driver-agnostic targeting of amplified surface proteins identifies MPZL1 for selective CAR-T cell therapy" paper that focuses on researching and identifying MPZL1 surface proteins to attack and destroy cancer tumors using CAR-T cells. The researchers used the mass spectrometry-derived Cell Surface Protein Atlas to identify 1,492 human cell-surface proteins and 2703 ICGC/TCGA pan-cancer samples to determine which of the proteins have extra copies in tumors by analyzing the number of copies using GISTIC 2.0 and cBioPortal. The researchers were able to narrow the list to 90 proteins. They then used GEPIA2, comparing gene activity in tumors versus healthy tissues, to determine which genes have a high presence in liver cancer. They used UniProt and Human Protein Atlas to identify 6 genes that are localized to the plasma membrane.

After screening through and narrowing down the list down to 6 genes, the researchers ultimately chose to target MPZL1 proteins since they do not affect tumor growth and have a higher presence on tumor cells than healthy tissue. The team then studied 2335 resected cancers across 6 different tumors and 30 normal tissues to understand MPZL1 protein expression in the tissues. Once understanding MPZL1's behavior, the team was able to construct a CAR-T cell that has a newly created antibody attached. This antibodies scFv grabbing piece locates the MPZL1 to latch onto the protein. Once it latches onto the protein, it signals to the T-cell to attack and kill. The CAR-T cells also had a LNGFR marker on them so that the researchers could identify them inside the tumor. The researchers tested the CAR-T cells in dishes, mice with human tumors, mice that grew liver tumors, and human tissue.

The paper focuses on a statistical analysis rather than a trained Machine Learning predictor. The researchers used MPZL1 KO cells to compare the CAR-T cells' behavior towards cells without the MPZL1 protein to cells with the MPZL1 protein present. The researchers focused their statistical analysis on showing that MPZL1 genes are the appropriate gene to target in tumors and understanding if the CAR-T cells attack cells because of the presence of the MPZL1 protein. The paper uses analysis of variance (ANOVA) to analyze how MPZL1 mRNA differs across normal liver, diploid HCC, and MPZL1-gain HCC and how tumor volume differs across days after CAR-T treatment. Tukey's multiple-comparisons test was then applied to produce adjusted p values. The log-rank test compares survival of mice with MPZL1 tumors and mice with MPZL1-KO tumors. The researchers also used correlation to analyze whether there is a positive relationship between MPZL1 gene copies and MPZL1 mRNA in tumors. The statistical metrics in their analysis are p-values, survival curves, and percent cell lysis. They found that at the highest E/T ratio utilized in the dishes, CAR-T cells killed 80% to 100% of all MPZL1 high target cancer cell lines. 7 out of 11 MPZL1 high tumors were eradicated and did not reappear in mice with human liver tumors. For mice where the tumors grow naturally, they did not have any tumor development and survived 100 days. MPZL1 CAR-T cells infiltrated MPZL1-positive tumor explants more than UT T cells in human tissue slices.

The paper does have unstated details that are very important for our team to know. For instance, the paper states that the researchers were able to successfully narrow the list of 1,492 cell-surface proteins down to 90 candidates, but we need to understand how the screening criteria were applied to create this final list. It is stated that the researchers used GISTIC2.0 in order to classify copy-number alterations and selected genes with copy-number gains or amplifications in more than 30% of samples. However, our

team needs to determine how this criterion was applied to the original dataset and how we can successfully reproduce the same 90 genes so that we can successfully reproduce the public analysis.

Our analysis will use a predictive machine learning model to predict a gene dependency score, which is trained using the negative median DepMap Chronos effect across at least three lung adenocarcinoma (LUAD) lines. A higher dependency score means the cells have a greater need for the gene. The unit of analysis we are studying is individual gene types. The methods we will use in our model include a constant reference to compare the other models to, ridge to add penalty weights so that we can develop a linear relationship and avoid overfitting to our data, random forest to analyze nonlinear relationships and interactions between features, and gradient boosting to evaluate if sequential combining improves accuracy. We will use a 70/15/15 split and ensure that each split includes surface eligible genes. The training split will include at least 20 surface eligible genes. Our held-out data is the 15% test set of genes that will not be used to train the model, nor used for model validation and improvement. The model output will be the predicted dependency score for each gene, and the metrics used for the level of performance will include Mean Absolute Error (MAE) and the Spearman Correlation.

The public analysis that our team will use to reproduce findings of the researchers includes the publicly available datasets, supplementary tables, and analysis tools that are referenced in the paper. In the paper it is stated that the researchers used the Cell Surface Protein Atlas and ICGC/TCGA pan-cancer data in order to identify amplified cell-surface protein genes, as well as with the cBioPortal and GEPIA2 in order to analyze copy-number alterations and gene expression. Our team will use the resources from the paper as well as the supplementary tables, and source data in order to investigate how the original list of 1,429 proteins was narrowed down to 90 candidates and then ultimately identified as MPZL1. We will compare our results to the reported results in order to determine whether we are able to reproduce the screening process and its results. For our separate machine learning task, we have also collected LUAD gene expression data from UCSC Xena, copy-number data from cBioPortal, protein localization annotations from Human Protein Atlas, and gene dependency data from DepMap. These different datasets will be combined into a gene-level table in order to train and evaluate our predictive models.

The implementation of the project will be in Jupyter Notebook with the use of Python, Pandas, and Scikit-learn. The general layout of the notebook will include loading and exploring public data, creating labels and organizing features, and establishing the training set. Effect-derived predictors will be excluded for data leakage. So far, we have downloaded UCSC Xena TCGA LUAD data that was used to collect the list of genes present in LUAD. The list of genes is used as the unit of analysis in Table 1: Gene Table. We used the gene expressions from this dataset to evaluate the expression mean and standard deviation for each gene. We downloaded TCGA LUAD GISTIC data from cBioPortal to determine the amplification frequency, which is the fraction of LUAD tumors with extra copies of the gene. We downloaded subcellular location annotations data from Human Protein Atlas to analyze if the genes appear on the surface of cells. We downloaded Chronos CRISPR gene effect data from DepMap to determine the negative median Chronos effect across the LUAD lines. Our ML table includes each of these features: gene, expression mean, expression standard deviation, amplification frequency, cell surface appearance, negative median Chronos effect across the LUAD lines. We have joined each of the datasets into a one row per gene table in Jupyter Notebook for 2000 genes removing duplicates. An example of this table is shown below in Table 1: Gene Table. The full data table can be found in our start-evidence package.

Table 1: Gene Table

Gene	Expression	Enrichment	Variability	Amplification	Dependency Score	Surface Level
NSUN5	9.334394	0.939807	0.658270	0.005814	0.164239	No
CUX2	1.856583	0.112388	1.096357	0.003876	0.123804	No
…	…	…	…	…	…	…
PCLO	7.141597	-0.110730	1.318136	0.015504	-0.078496	Yes

For Checkpoint 1, two group members worked on the public paper analysis (Caroline Nutter & Joanna Land) and two group members worked on preparing the start evidence package (Skyler Skinner & Nadia Alexander) for our machine learning model. The next step of this project is to start working on reproducing the public statistical analysis and implementing code for our machine learning model to build the constant/Ridge baselines by Checkpoint 2. Caroline and Joanna will work on the public statistical analysis, and Skyler and Nadia will work on model implementation. We will work to have a working replicated statistical analysis and model by Checkpoint 2. For Checkpoint 3, we as a group will plan to perform model improvements together in order to complete this project.

Citation

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